

DATEX-OHMEDA COM OUTPUT PROTOCOLS FOR CARESTATION 750

DATEX-OHMEDA COM 1.2 SERIAL PROTOCOL

DATEX-OHMEDA COM 1.0 SERIAL PROTOCOL

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CONTENTS

1. PURPOSE	4
2. OHMEDA COM 1.0 SERIAL PROTOCOL	4
2.1 ELECTRICAL INTERFACE	4
2.2 SOFTWARE INTERFACE	4
2.3 SUMMARY OF INPUT COMMANDS AND OUTPUT DATA.....	5
2.4 DEVICE COMMANDS	7
2.4.1 Auto Mode	7
2.4.2 Slave Mode	7
2.4.3 Send All Data	7
2.4.4 Send Setup Data	7
2.4.5 Disable Checksum Control Command	7
2.4.6 Enable Checksum Control Command	8
2.4.7 Enable Waveform Data Mode Command	8
2.5 DEVICE RESPONSES	8
2.5.1 ACK Response.....	9
2.5.2 NACK Response	9
2.5.3 Alarm Silence Switch Pressed Response.....	9
2.5.4 Measured Data Response	9
2.5.5 Status Data Response	10
2.5.6 Setup Data Response	16
2.5.7 Waveform Data Response	17
2.6 EXAMPLE	19
3. DATEX-OHMEDA COM 1.2 SERIAL PROTOCOL	21
3.1 ELECTRICAL INTERFACE	21
3.2 SOFTWARE INTERFACE	21
3.3 SUMMARY OF INPUT COMMANDS AND OUTPUT DATA.....	22
3.4 DEVICE COMMANDS	23
3.4.1 Auto Mode	23
3.4.2 Slave Mode	23
3.4.3 Datex-Ohmeda Communications Protocol Command	23
3.4.4 Send All Data	23
3.4.5 Send Setup Data	24
3.4.6 Disable Checksum Control Command	24
3.4.7 Enable Checksum Control Command	24
3.4.8 Enable Waveform Data Mode Command	24
3.4.9 Enable Privileged Mode	27
3.4.10 Silence Alarms Command	27
3.4.11 Synchronize Clock Command	27
3.4.12 Synchronize Demographic Data Command	27
3.5 DEVICE RESPONSES	28
3.5.1 ACK Response.....	28

3.5.2	NACK Response	29
3.5.3	Alarm Silence Switch Pressed Response.....	29
3.5.4	Measured Data Response	29
3.5.5	Status Data Response	35
3.5.6	Setup Data Response	47
3.5.7	Waveform Data Response	48
3.5.8	Test Data Response	53
3.6	EXAMPLE	58
4.	CHANGE HISTORY	60

1. Purpose

The purpose of this document is to describe and give examples for the following Datex-Ohmeda Com output protocols as it applies to the Carestation 750 Anesthesia workstation:

Datex-Ohmeda Com 1.2 Protocol
Datex-Ohmeda Com 1.0 Protocol

1.1 Security Recommendations

GE Healthcare recommends organizations use secure terminal servers if choosing to connect GE Healthcare anesthesia device serial ports to TCP/IP networks. Secure terminal servers, when correctly configured, provide robust security features, including strong encryption, VPN, authentication of users, network controls, logging, audit capability, and secure device configuration and management options.

GE Healthcare recommends that organizations utilize best practices for terminal servers that include governance, management and secure deployment measures such as network segmentation, VLANs and device isolation to enhance existing security measures.

2. Ohmeda Com 1.0 Serial Protocol

The following section describes the Ohmeda Com 1.0 Protocol. This protocol is an available protocol to communicate with the Datex-Ohmeda Carestation 750 Anesthesia workstations.

2.1 Electrical Interface

RS-232C signal standards:

15 pin female D connector – Data Communications Equipment configuration (DCE)

pin 6 - receive data
pin 13 - transmit data
pin 5 - signal ground

Baud Rate: 19.2K
Byte format: Start bit + 7 data bits + parity bit + stop bit
Parity: ODD

2.2 Software Interface

Ohmeda Com 1.0 Communications Protocol

Notes:

1. All device commands (sent by communications controller) begin with the ASCII escape character (0x1b).
2. All device responses (sent by Datex-Ohmeda device) begin with the ASCII character “:”.
3. The next two bytes are the device designator. “VT” designates a ventilator. Rightly or wrongly, the Datex-Ohmeda Carestation 750 Anesthesia workstation will use the “VT” designator.
4. All parameter fields consist of byte-wise 7 bit printable ASCII characters. All status fields consist of 7 bit bytes that may or may not be printable characters.
5. All numeric parameter fields are ASCII encoded decimals, right justified, with leading digits zeroed.
6. All waveform data fields are 3 digits of ASCII encoded hexadecimal (7 bit printable characters), right justified, with leading digits zeroed. If the requested waveform data is not available, data is zeroed.
7. The checksum byte for the D-O communications protocol is defined as a 7-bit number which, when added to the accumulated sum of all the bytes transmitted in a command or response (does not include the carriage return character following the checksum byte), causes the least significant 7 bits of the result to be zero. The checksum may or may not be a printable character.
8. All messages end with the ASCII carriage return (0x0d).
9. Use Auto Mode whenever possible. In Slave Mode, do not request Send All Data more than once per second.
10. Do not command Enable Waveform Data more than once per second.

2.3 Summary of Input Commands and Output Data

When the Datex-Ohmeda device is first turned ON, the default transmission mode is set to “Slave Mode”, the Checksum Mode is “Enabled”, and Waveforms are disabled.

Command Headers	Description
<ESC>VTD	Disable Checksum Mode
<ESC>VTE	Enable Checksum Mode (this is the default mode)
<ESC>VTS	Slave Mode (this is the default mode – will reset auto mode)
<ESC>VTX	Auto Mode (see description of Measured Data and Status Data Responses)
<ESC>VTQ	Enable Compressed Data Format (this is the default mode – since printer data format is no longer supported, command is simply ACK’ed)
<ESC>VT\$	Send Setup Data
<ESC>VT?	Send All Data (will send one Status and Measured Data Response – not valid in Auto Mode)
<ESC>VTW	Enable Waveform Mode (default is no waveforms enabled)
<ESC>VTM	NOT USED - Enable Privileged Mode (must be sent before using VTCS and VT% commands)
<ESC>VTCS	NOT USED - Silence Alarms (simulate alarm silence switch press, requires privileged mode)

<ESC>VT%	NOT USED - Gas Composition Input
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Response Headers	Description
:VTD	Measured Data Response
:VTM	Setup Data Response
:VTN	NACK (negative acknowledgement)
:VTQ	Status Data Response
:VTR	Alarm Silence Switch Press Response
:VTW	Waveform Data Response
:VTY	ACK (positive acknowledgement)

2.4 Device Commands

2.4.1 Auto Mode

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTX
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

2.4.2 Slave Mode

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTS
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

2.4.3 Send All Data

Valid in Slave Mode Only

Byte(s)	Field Name	Description
0-3	Header	<ESC>VT?
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

2.4.4 Send Setup Data

Byte(s)	Field Name	Description
0-3	Header	<ESC>VT\$
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

2.4.5 Disable Checksum Control Command

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTD
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

2.4.6 Enable Checksum Control Command

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTE
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

2.4.7 Enable Waveform Data Mode Command

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTW
4	Any one of the following: (note 1: order determines order of data in response packet)	'0' Turn waveform data OFF (this is the ASCII character <u>zero</u>)
		'P' Include Pressure Data
		'F' Include Flow Data
		'V' Include Volume Data
5	Any one of the following: (note 1: order determines order of data in response packet)	'0' Turn waveform data OFF (this is the ASCII character <u>zero</u>)
		'P' Include Pressure Data
		'F' Include Flow Data
		'V' Include Volume Data
6	Checksum	See Software Interface section.
7	Carriage Return	<CR>

2.5 Device Responses

2.5.1 ACK Response

This response applies to the commands of VTD, VTE, VTS, VTX, VTQ, VTW, VTM, VTCS, and VT%.

Byte(s)	Field Name	Description
0-3	Header	:VTY
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

2.5.2 NACK Response

Byte(s)	Field Name	Description
0-3	Header	:VTN
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

2.5.3 Alarm Silence Switch Pressed Response

Byte(s)	Field Name	Description
0-3	Header	:VTR
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

2.5.4 Measured Data Response

In auto mode, the Measured Data Response will be transmitted at the end of a breath or 10 seconds from the last transmission, whichever occurs first.

Each entry is zero filled and right justified – i.e. 95 expressed in four bytes is 0095.

“?” means bad data due to any technical problem(s).

“-” filled means data not available due to system state.

Byte(s)	Measured Data Name	Units and/or Description
0-3	Header	:VTD
4-7	Measured expired tidal volume	mL, ?, -
8-11	Measured expired minute volume	L*100, ?, -
12-14	Measured respiratory rate	/min, ?, -
15-17	Measured oxygen level	% O ₂ , ?, -
18-20	Measured max positive pressure	cmH ₂ O, ?, -
21-23	Measured inspiratory plateau pressure	cmH ₂ O, ?, -
24-26	Measured mean pressure	cmH ₂ O, ?, -
27-29	Not Used	-
30	Measured data status	See section 2.5.4.1
31	Checksum	See Software Interface section.
32	Carriage return	<CR>

2.5.4.1 Status Bytes Bitmaps

The status byte has seven bits of data from D6 (MSB) to D0 (LSB).

Byte 30	
Bit	Alarm/Status
D0	0 = 10 Second Data 1 = New Breath Data
D1	Reserved
D2	Reserved
D3	Reserved
D4	Reserved
D5	Reserved
D6	Reserved

2.5.5 Status Data Response

In auto mode, the Status Data Response will be transmitted every second (if a change occurs in status data) or a minimum of every 10 seconds.

Note that the status byte bit is set = 1 for an active condition and 0 for an inactive condition.

Each entry is zero filled and right justified – i.e. 95 expressed in four bytes is 0095.

Byte(s)	Status Data Name	Units and/or Description	
0-3	Header	:VTQ	
4-7	Set tidal volume	mL	
8-10	Set mechanical respiration rate	/min	
11-14	Set I:E	1:eee.e	
15-16	Set Inspiratory pause %	% Pause	
17-18	Set PEEP	cmH ₂ O	
19-21	Set peak pressure limit	cmH ₂ O	
22-23	Set inspired pressure	cmH ₂ O	
24-25	Not Used	-	
26-28	Set High MVexp alarm limit	L * 10	
29-31	Set Low MVexp alarm limit	L * 10	
32-34	Set High TVexp alarm limit	mL / 10	
35-37	Set Low TVexp alarm limit	mL / 10	
38-40	Set High O ₂ alarm limit	% O ₂	
41-43	Set Low O ₂ alarm limit	% O ₂	
44	Set ventilation mode	'v'	Volume control mode (VCV, CMV)
		'p'	Pressure control mode (PCV)
		'p'	PCV-VG
		'p'	BiLevel-VG
		'p'	SIMV-VC
		'p'	SIMV-PC
		'p'	SIMV-PCVG
		'p'	BiLevel
		'p'	CPAP/PSV
		'p'	NIV

		'-'	In bag mode or mechanical ventilation not available Note: Please see section 3.5.5.1 byte 11, bit D6 'Mechanical Ventilation On' for determining whether system is mechanically ventilating or not.
45-56	Status bytes		
57	Checksum	See Software Interface section.	
58	Carriage return	<CR>	

2.5.5.1 Status Bytes Bitmaps

The status bytes a string of 12 bytes, starting from left (byte 1) to right (byte 12). Each byte has seven bits of data from D6 (MSB) to D0 (LSB).

Note that the status byte bit is set = 1 for an active condition and 0 for an inactive condition.

Byte 45	
Bit	Alarm/Status
D0	High O ₂
D1	Low O ₂
D2	Reserved
D3	Reserved
D4	Reserved
D5	Replace O ₂ Sensor
D6	Calibrate O ₂ Sensor

Byte 46	
Bit	Alarm/Status
D0	High Paw
D1	Low Paw
D2	Not Used
D3	Sustained Paw
D4	Paw < -10 cmH ₂ O
D5	Not Used
D6	Reserved

Byte 47	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Pressure Mode not available
D3	Manifold Pressure Sensor Failure, Minimum System
D4	Not Used
D5	Inspiration Stopped !
D6	Not Used

Byte 48	
Bit	Alarm/Status
D0	Low V_E
D1	High V_E
D2	Low V_{TE}
D3	High V_{TE}
D4	V_T Not Achieved
D5	Volume Apnea
D6	Volume Apnea > 2 min

Byte 49	
Bit	Alarm/Status
D0	No Inspiratory Flow Sensor
D1	No Expiratory Flow Sensor
D2	Inspiratory Reverse Flow
D3	Expiratory Reverse Flow
D4	Check Flow Sensors
D5	$V_{TE} > \text{Insp } V_T$
D6	System Leak ? or Patient Circuit Leak

Byte 50	
Bit	Alarm/Status
D0	Cannot Drive Bellows
D1	Flow Valve (Current) Failure, Flow Valve (DAC) Failure, minimum system
D2	Not Used
D3	12 Hour Test !
D4	Bootup Gas Inlet Valve Failure
D5	No O_2 Pressure
D6	Not Used

Byte 51	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 52	
Bit	Alarm/Status
D0	Not Used
D1	CPU Failure
D2	Memory (EEPROM) Failure
D3	Memory (Flash) Failure
D4	Memory (RAM) Failure
D5	Not Used
D6	Not Used

Byte 53	
Bit	Alarm/Status
D0	Software Watchdog Failure
D1	Hardware Watchdog Failure
D2	Internal Clock Too Fast
D3	Not Used
D4	CPU Internal Error
D5	Memory (redundant storage) Fail
D6	Insp Flow Sensor Failure, Exp Flow Sensor Failure

Byte 54	
Bit	Alarm/Status
D0	On Battery – Power OK ?
D1	Battery Failure Low
D2	Not Used
D3	Low Battery Voltage
D4	Not Used
D5	Not Used
D6	Not Used

Byte 55	
Bit	Alarm/Status
D0	Not Used
D1	Auxiliary Breathing Circuit
D2	Not Used
D3	Control Settings Input Has Failed
D4	Not Used
D5	Backup Volume Mode
D6	Mechanical Ventilation ON

Byte 56	
Bit	Alarm/Status
D0	Not Used
D1	Apnea Standby = OFF
D2	Not Used
D3	Not Used
D4	Not Used
D5	Reserved
D6	Service Calibrations !

2.5.6 Setup Data Response

Only returned when a request for setup data is received.

Each entry is zero filled and right justified – i.e. 95 expressed in four bytes is 0095.

Byte(s)	Setup Data Name	Units and/or Description	
0-3	Header	:VTM	
4-5	Software revision number	01-99	
6-7	Service Pack Number	00-99	
8	Language	'0'=English, '1'=Spanish, '2'=German, '3'=Kanji, '4'=Dutch, '5'=Swedish, '6'=French, '7'=Italian, '8'=Danish, '9'=Polish, 'A'=Chinese, 'B'=Czech, 'C'=Hungarian, 'D'=Russian, 'E'=Portuguese, 'F'=Other	
9-10	Display contrast setting	1-64	
11	Alarm volume setting	1-5	
12-13	Altitude setting	-4 to 36 in 100's of meters	
14	Drive gas	'O'	Oxygen
		'A'	Air
15	Model number	'6' = depicts 7900 anesthesia workstation.	
16	Checksum	See Software Interface section.	
17	Carriage return	<CR>	

2.5.7 Waveform Data Response

If Waveform Data Mode is enabled, a Waveform Data Response will be transmitted every 240 mS. Each data value is a 3-digit, zero filled, right justified ASCII Hex representation of a 12 bit binary value.

Each entry is zero filled and right justified – i.e. 95 expressed in three bytes is 095.

If the requested waveform data is not available, data is zeroed.

For each waveform signal selected (maximum of two signals).

Byte(s)	Waveform Data Name	Units and/or Description
0-3	Header	:VTW
4-6	1st 16ms waveform sample	hexadecimal "000" - "FFF" (see section 2.5.7.1 Waveform Data Description)
7-9	2nd 16ms waveform sample	hexadecimal "000" - "FFF" (see section 2.5.7.1 Waveform Data Description)
.	.	.
.	.	.
.	.	.
43-45	14th 16ms waveform sample	hexadecimal "000" - "FFF" (see section 2.5.7.1 Waveform Data Description)
46-48	15th 16ms waveform sample	hexadecimal "000" - "FFF" (see section 2.5.7.1 Waveform Data Description)
49	Checksum	
50	Carriage return	<CR>

2.5.7.1 Waveform Data Description

Each waveform data sample shall be scaled as follows:

Pressure		
Range: -20 to 120 cmH ₂ O		
Raw Data	Scaled value	Transmitted
-20	0	"000"
0	512	"200"
120	3584	"E00"

Flow		
Range: -100 to 100 L/min		
Raw Data	Scaled value	Transmitted
-100	512	"200"
0	2048	"800"
100	3584	"E00"

Volume		
Range: 0 to 2000 mL		
Raw Data	Scaled value	Transmitted
0	512	"200"
2000	3584	"E00"

2.6 Example

The Carestation 750 Anesthesia Workstations default serial protocol is version 1.0 in "Slave Mode" with "Checksum Enabled". So data **will not** automatically be sent out the serial port.

If an external communications controller wants the Datex-Ohmeda device to stay in "Slave Mode" it must send the <ESC>VT?<checksum><CR> packet ("Send All Data") to receive one Status Data and one Measured Data packet. The external communications controller will then need to request sending all the data on a periodic basis. This method is not recommended because the external communications controller has the potential to request sending all the data faster than the Datex-Ohmeda device is capable of responding.

So, the preferred method is to use "Auto Mode" by sending the <ESC>VTX<checksum><CR> packet. This will allow the Datex-Ohmeda device to send the measured data and status data whenever changes occur, or every 10 seconds when changes do not occur. Using this method **will not** put a strain on the Datex-Ohmeda device and again is the preferred method to use.

Listed below is an example to get a Datex-Ohmeda device into auto mode with no checksum and pressure waveform enabled. Using this method the <checksum> can be any character (including carriage return, <CR>) for all commands sent after the <ESC>VTDw<CR> command (in this example space, <SP>, is used).

<ESC>VTDw<CR>	- Disable checksum ("w" or 0x77 is the checksum value for this command)
<ESC>VTX<SP><CR>	- Set into Auto mode (<SP> used for checksum since checksum is disabled)

<ESC>VTWP0<SP><CR>

- Request pressure waveform data, no second waveform

3. Datex-Ohmeda Com 1.2 Serial Protocol

The following section describes the Datex-Ohmeda Com 1.2 Protocol. This protocol is an available protocol to communicate with the Datex-Ohmeda Carestation 750 Anesthesia workstations.

3.1 Electrical Interface

RS-232C signal standards:

15 pin female D connector – Data Communications Equipment configuration (DCE)

- pin 6 - receive data
- pin 13 - transmit data
- pin 5 - signal ground

Baud Rate: 19.2K
Byte format: Start bit + 7 data bits + parity bit + stop bit
Parity: ODD

3.2 Software Interface

Datex-Ohmeda Com 1.2 Communications Protocol

Notes:

1. All device commands (sent by communications controller) begin with the ASCII escape character (0x1b).
2. All device responses (sent by Datex-Ohmeda device) begin with the ASCII character “:”.
3. The next two bytes are the device designator. “VT” designates a ventilator. Rightly or wrongly, the Datex-Ohmeda Carestation 750Anesthesia workstations will use the “VT” designator.
4. All parameter fields consist of byte-wise 7 bit printable ASCII characters. All status fields consist of 7 bit bytes that may or may not be printable characters.
5. All numeric parameter fields are ASCII encoded decimals, right justified, with leading digits zeroed.
6. All waveform data fields are 3 digits of ASCII encoded hexadecimal (7 bit printable characters), right justified, with leading digits zeroed. If the requested waveform data is not available, data is zeroed.
7. The checksum byte for the D-O communications protocol is defined as a 7-bit number which, when added to the accumulated sum of all the bytes transmitted in a command or response (does not include the carriage return character following the checksum byte), causes the least significant 7 bits of the result to be zero. The checksum may or may not be a printable character.
8. All messages end with the ASCII carriage return (0x0d).
9. Use Auto Mode whenever possible. In Slave Mode, do not request Send All Data more than once per second.

10. Do not command Enable Waveform Data more than once per second.

3.3 Summary of Input Commands and Output Data

When the Datex-Ohmeda device is first turned ON, the default transmission mode is set to "Slave Mode", the Checksum Mode is "Enabled" and Waveforms are disabled.

Command Headers	Description
<ESC>VTD	Disable Checksum Mode
<ESC>VTE	Enable Checksum Mode (this is the default mode)
<ESC>VTS	Slave Mode (this is the default mode – will reset auto mode)
<ESC>VTX	Auto Mode (see description of Measured Data and Status Data Responses)
<ESC>VTQ	Enable Compressed Data Format (this is the default mode – since printer data format is no longer supported, command is simply ACK'ed)
<ESC>VT\$	Send Setup Data
<ESC>VT?	Send All Data (will send one Status and Measured Data Response – not valid in Auto Mode)
<ESC>VTO	Select Datex-Ohmeda Com Protocol
<ESC>VTw	Enable Waveform Mode (default is no waveforms enabled)
<ESC>VTCC	NOT USED - Synchronize Real Time Clock
<ESC>VTM	NOT USED - Enable Privileged Mode (must be sent before using VTCS command)
<ESC>VTCS	NOT USED - Silence Alarms (simulate alarm silence switch press)
<ESC>VTCD	NOT USED - Synchronize Demographic Data

Response Headers	Description
:VTd	Measured Data Response
:VTm	Setup Data Response
:VTN	NACK (negative acknowledgement)
:VTq	Status Data Response
:VTR	Alarm Silence Switch Press Response
:VTt	Test Data Response
:VTw	Waveform Data Response
:VTY	ACK (positive acknowledgement)

3.4 Device Commands

3.4.1 Auto Mode

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTX
4	Checksum	Checksum
5	Carriage return	<CR>

3.4.2 Slave Mode

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTS
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

3.4.3 Datex-Ohmeda Communications Protocol Command

Note: This command must be used to enable the Datex-Ohmeda Com 1.2 protocol.

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTO
4-5	Requested Protocol Version	"10" Ohmeda Com 1.0 protocol (See section 2)
		"12" Ohmeda Com 1.2 protocol
6	Checksum	See Software Interface section.
7	Carriage return	<CR>

3.4.4 Send All Data

Valid in Slave Mode Only

Byte(s)	Field Name	Description
0-3	Header	<ESC>VT?
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

3.4.5 Send Setup Data

Byte(s)	Field Name	Description
0-3	Header	<ESC>VT\$
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

3.4.6 Disable Checksum Control Command

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTD
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

3.4.7 Enable Checksum Control Command

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTE
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

3.4.8 Enable Waveform Data Mode Command

Note 1: Order of activated waveform selections determines order of data in response packet.

Note 2: Do not duplicate activated waveform selections.

Note 3: Use ASCII character zero to deactivate a waveform.

Note 4: Undefined selections are treated as ASCII character zero.

Note 5: If Waveform 1 is set to '0', all waveform will be disabled.

Byte(s)	Field Name	Description	
0-3	Header	<ESC>VTw	
4	Waveform 1 Any one of the following: (note: order determines order of data in response packet)	'0'	Turn waveform data OFF (this is the ASCII character <u>zero</u>)
		'P'	Include Pressure Data
		'F'	Include Flow Data
		'V'	Include Volume Data
		'X'	Include Auxiliary Pressure Data
		'A'	Include Anesthetic Agent Data
		'C'	Include CO ₂ Data
		'O'	Include O ₂ Data
5	Waveform 2 Any one of the following: (note: order determines order of data in response packet)	'0'	Turn waveform data OFF (this is the ASCII character <u>zero</u>)
		'P'	Include Pressure Data
		'F'	Include Flow Data
		'V'	Include Volume Data
		'X'	Include Auxiliary Pressure Data
		'A'	Include Anesthetic Agent Data
		'C'	Include CO ₂ Data
		'O'	Include O ₂ Data
6	Waveform 3 Any one of the following: (note: order determines order of data in response packet)	'0'	Turn waveform data OFF (this is the ASCII character <u>zero</u>)
		'P'	Include Pressure Data
		'F'	Include Flow Data
		'V'	Include Volume Data
		'X'	Include Auxiliary Pressure Data
		'A'	Include Anesthetic Agent Data
		'C'	Include CO ₂ Data
		'O'	Include O ₂ Data

7	Waveform 4 Any one of the following: (note: order determines order of data in response packet)	'0'	Turn waveform data OFF (this is the ASCII character <u>zero</u>)
		'P'	Include Pressure Data
		'F'	Include Flow Data
		'V'	Include Volume Data
		'X'	Include Auxiliary Pressure Data
		'A'	Include Anesthetic Agent Data
		'C'	Include CO ₂ Data
		'O'	Include O ₂ Data
8	Waveform 5 Any one of the following: (note: order determines order of data in response packet)	'0'	Turn waveform data OFF (this is the ASCII character <u>zero</u>)
		'P'	Include Pressure Data
		'F'	Include Flow Data
		'V'	Include Volume Data
		'X'	Include Auxiliary Pressure Data
		'A'	Include Anesthetic Agent Data
		'C'	Include CO ₂ Data
		'O'	Include O ₂ Data
9	Waveform 6 Any one of the following: (note: order determines order of data in response packet)	'0'	Turn waveform data OFF (this is the ASCII character <u>zero</u>)
		'P'	Include Pressure Data
		'F'	Include Flow Data
		'V'	Include Volume Data
		'X'	Include Auxiliary Pressure Data
		'A'	Include Anesthetic Agent Data
		'C'	Include CO ₂ Data
		'O'	Include O ₂ Data
10	Checksum	See Software Interface section.	
11	Carriage return	<CR>	

3.4.9 Enable Privileged Mode

Byte(s)	Field Name	Description
0-3	Header	<ESC>VTM
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

3.4.10 Silence Alarms Command

Note: In Privileged Mode only.

Byte(s)	Field Name	Description
0-4	Header	<ESC>VTCS
5	Checksum	See Software Interface section.
6	Carriage return	<CR>

3.4.11 Synchronize Clock Command

Each entry is zero filled and right justified – i.e. 95 expressed in four bytes is 0095.

Byte(s)	Field Name	Units and/or Description
0-4	Header	<ESC>VTCC
5-12	Date (yyyymmdd)	Year, month, day (example: 20031231)
13-16	Time (hhmm)	Hour (24 hour format), minutes (example: 2359)
17	Checksum	See Software Interface section.
18	Carriage return	<CR>

3.4.12 Synchronize Demographic Data Command

Each entry is zero filled and right justified – i.e. 95 expressed in four bytes is 0095.

Byte(s)	Field Name	Units and/or Description	
0-4	Header	<ESC>VTCD	
5	Set patient type	'a'	Adult
		'p'	Pediatric
		'b'	Ideal body weight (IBW)
6-8	Set IBW Set patient ideal body weight	Kg, -	
9-11	Set BSA Set patient body surface area	m ² *100, -	
12-14	Set patient weight	Kg, -	
15-17	Set patient height	cm, -	
18-27	Set patient ID	(limited to 7-bit ASCII characters)	
28-32	Set patient bed location	(limited to 7-bit ASCII characters)	
33-40	Set patient age in years, months, and days (yyyymmdd)	(use leading zeroes and zeroes that are significant, months or days fields that are not significant are dashed) (example: 0002000-- = 2 years and 0 months, days are not significant)	
41	Checksum	See Software Interface section.	
42	Carriage return	<CR>	

3.5 Device Responses

3.5.1 ACK Response

This response applies to the commands of VTD, VTE, VTS, VTX, VTQ, VTO, VTW, VTM, VTCS, VTCC, and VTCD.

Byte(s)	Field Name	Description
0-3	Header	:VTY
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

3.5.2 NACK Response

Byte(s)	Field Name	Description
0-3	Header	:VTN
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

3.5.3 Alarm Silence Switch Pressed Response

Byte(s)	Field Name	Description
0-3	Header	:VTR
4	Checksum	See Software Interface section.
5	Carriage return	<CR>

3.5.4 Measured Data Response

In auto mode, the Measured Data Response will be transmitted at the end of a breath (but not more than once every 0.5 seconds) or 10 seconds from the last transmission, whichever occurs first.

Each entry is zero filled and right justified – i.e. 95 expressed in four bytes is 0095.

“?” means bad data due to any technical problem(s).

“-“ filled means data not available due to system state.

Byte(s)	Measured Data Name	Units and/or Description
0-3	Header	:VTd
4-7	TVexp Measured expiratory tidal volume (mechanical or spontaneous)	mL, ?, -
8-11	MVexp Measured total expiratory minute volume (mechanical or spontaneous)	L*100, ?, -
12-14	RRtotal Measured spirometry total respiratory rate (mechanical + spontaneous)	/min, ?, -
15-17	Circuit O ₂ (internal) Measured circuit oxygen conc.	% O ₂ (volume), ?, -
18-20	Ppeak Measured max positive airway pressure	cmH ₂ O, ?, -
21-23	Pplat Measured inspiratory plateau airway pressure (requires inspiratory pause)	cmH ₂ O, ?, -
24-26	Pmean Measured mean pressure	cmH ₂ O, ?, -
27-29	Not Used	-
30-33	MVexp spont measured spontaneous expiratory minute volume	L*100, ?, -
34-36	RR spont measured spontaneous respiratory rate	/min, ?, -
37-39	PEEPi measured intrinsic PEEP	cm H ₂ O*10, ?, -

40-41	compliance measured compliance	mL/cm H ₂ O, ?, -	
42-44	Raw measured airway resistance	(cm H ₂ O/L/s)*10, ?, -	
45	Punits Displayed pressure units	'm'	Mbar
		'c'	cmH ₂ O
		'k'	kPa
		'h'	hPa
		'g'	mmHg
46	Funits Displayed flow units	'm'	L/min
		's'	L/s
		'c'	mL/s
47-50	TVexp spont measured spontaneous expiratory tidal volume	mL, ?, -	
51-54	TVinsp measured inspiratory tidal volume (mechanical or spontaneous)	mL, ?, -	
55-58	MVinsp measured inspiratory minute volume (mechanical or spontaneous)	L*100, ?, -	
59-61	Not Used	-	
62-64	Not Used	-	
65-67	Not Used	-	
68-70	PEEPe measured extrinsic PEEP	cm H ₂ O*10, ?, -	
71-73	PEEPe+i measured total PEEP (intrinsic + extrinsic)	cm H ₂ O*10, ?, -	
74-75	Not Used	-	
76-77	Not Used	-	

78-79	Not Used	-	
80-82	measured ambient pressure	mm Hg, ?, -	
83-85	FiO ₂ (MGAS) measured Fi oxygen conc.	% O ₂ (volume), ?, -	
86-88	etO ₂ measured end tidal oxygen conc.	% O ₂ (volume), ?, -	
89-92	FiO ₂ – etO ₂ measured oxygen conc. difference (FiO ₂ – etO ₂)	% O ₂ *10 (volume), ?, -	
93-95	FiCO ₂ measured Fi CO ₂ conc.	% CO ₂ *10 (volume), ?, -	
96-98	etCO ₂ measured end tidal CO ₂ conc.	% CO ₂ *10 (volume), ?, -	
99-101	RRCO ₂ measured CO ₂ total respiratory rate (mechanical + spontaneous)	/min, ?, -	
102-104	FiAA measured Fi anesthetic agent conc.	% AA*10 (volume), ?, -	
105-107	etAA measured end tidal anesthetic agent conc.	% AA*10 (volume), ?, -	
108	AA id Identified anesthetic agent	'0'	Not Used
		'1'	Isoflurane
		'2'	Not Used
		'3'	No agent
		'5'	Desflurane
		'6'	Sevoflurane
109-111	FiAA 2 nd measured secondary Fi anesthetic agent conc.	% AA*10 (volume), ?, -	

112-114	etAA 2 nd measured secondary end tidal anesthetic agent conc.	% AA*10 (volume), ?, -	
115	AA id 2 nd Measured secondary end tidal anesthetic agent conc.	'0'	Not Used
		'1'	Isoflurane
		'2'	Not Used
		'3'	No agent
		'5'	Desflurane
		'6'	Sevoflurane
116-118	FiN ₂ O measured Fi N ₂ O conc.	% N ₂ O*10 (volume), ?, -	
119-121	etN ₂ O measured end tidal N ₂ O conc.	% N ₂ O*10 (volume), ?, -	
122-123	MAC	no units*10, ?, -	
124-127	Not Used	-	
128-131	Not Used	-	
132-135	Not Used	-	
136-139	Not Used	-	
140-143	Not Used	-	
144-146	Not Used	-	
147	CO ₂ units	'%'	% CO ₂ (volume)
	Displayed CO ₂ units	'k'	kPa
		'g'	mm Hg
148-150	O ₂ supply pres measured oxygen supply pressure	kPa, ?, -	
151-153	N ₂ O supply pres measured N ₂ O supply pressure	kPa, ?, -	
154-156	air supply pres measured air supply pressure	kPa, ?, -	

157-159	O ₂ cylinder pres measured oxygen cylinder pressure	kPa*0.01, ?, -	
160-162	O ₂ cylinder pres 2 nd measured secondary oxygen cylinder pressure	kPa*0.01, ?, -	
163-165	N ₂ O cylinder pres measured N ₂ O cylinder pressure	kPa*0.01, ?, -	
166-168	air cylinder pres measured air cylinder pressure	kPa*0.01, ?, -	
169-171	Not Used	-	
172-174	Not Used	-	
175-177	Not Used	-	
178-180	Not Used	-	
181-183	Not Used	-	
184-187	O ₂ measured flow measured oxygen flow rate	L/min*100, ?, -	
188-191	N ₂ O measured flow measured N ₂ O flow rate	L/min*100, ?, -	
192-195	air measured flow measured air flow rate	L/min*100, ?, -	
196	Airway pressure measurement source	'i'	Internal
		'm'	MGAS
197	Flow measurement source	'i'	Internal
		'm'	MGAS
198	O ₂ measurement source	'i'	Internal
		'm'	MGAS
199-201	inspiratory time measured inspiratory time	s*10, ?, -	
202-204	expiratory time measured expiratory time	s*10, ?, -	
205-208	measured I:E ratio	ratio (1:xx.xx), ?, -	

209-211	Not Used	-
212-213	Measured data status	See section 3.5.4.1
214	Checksum	See Software Interface section.
215	Carriage return	<CR>

3.5.4.1 Status Bytes Bitmaps

The measured data status bytes are a string of 2 bytes, starting from left (byte 1) to the right (byte 2). Each byte has seven bits of data from D6 (MSB) to D0 (LSB).

Note that a bit is set to 1 for an active condition (true) and to 0 for an inactive condition (false).

Byte 212	
Bit	Alarm/Status
D0	0 = 10 second data refresh 1 = new breath data
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 213	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Reserved
D3	Reserved
D4	Reserved
D5	Reserved
D6	Reserved

3.5.5 Status Data Response

In auto mode, the Status Data Response will be transmitted every status change (but not more than once every second) or 10 seconds from the last transmission, whichever occurs first.

Each entry is zero filled and right justified – i.e. 95 expressed in four bytes is 0095.

‘-’ filled means data not available due to a system state or configuration.

Byte(s)	Status Data Name	Units and/or Description
0-3	Header	:VTq
4-7	set TV set tidal volume	mL, -
8-10	set RR set respiratory rate for controlled ventilation modes	/min, -
11-14	set I:E set I:E ratio	1:eee.e, - Example: Carestation 750 Anesthesia inverted I:E ratios are 2.0:1 = 1:000.5 and 1.5:1 = 1:000.6
15-16	set Tpause set inspiratory pause	% inspiratory time, -
17-18	set PEEP	cm H ₂ O, -
19-21	set Pmax set peak pressure limit (For Aestiva, mechanical breath cycles to exhalation if patient airway pressure exceeds P _{limit} . For Prism, P _{limit} is the limiting pressure, without cycling to exhalation, for a controlled volume breath and P _{max} is the pressure at which a mechanical breath cycles to exhalation.)	cm H ₂ O, -
22-23	set P _{insp} set inspired pressure	cm H ₂ O, -
24-25	Not Used	-
26-28	set High MVexp alarm set high expired minute volume alarm limit	L*10, -
29-31	set Low MVexp alarm set low expired minute volume alarm limit	L*10, -

32-35	set High TVexp alarm set high Vte alarm limit	ml*0.1, -	
36-39	set Low TVexp alarm set low Vte alarm limit	ml*0.1, -	
40-42	set High FiO ₂ alarm set high Fi oxygen alarm limit	% O ₂ (volume), -	
43-45	set Low FiO ₂ alarm set low Fi oxygen alarm limit	% O ₂ (volume), -	
46	Set mechanical ventilation mode	'v'	Volume control mode (VCV, CMV)
		'p'	Pressure control mode (PCV)
		'b'	Backup volume control mode
		'g'	PCV-VG
		'G'	BiLevel-VG
		's'	SIMV, SIMV-VC
		'i'	SIMV-PC
		'S'	SIMV-PCVG
		'B'	BiLevel
		'c'	CPAP, CPAP/PSV
		'a'	CPAP/Apnea
		'n'	NIV
		'o'	PSV-Pro

		'-'	In bag mode or mechanical ventilation not available Note: Please see section 3.5.5.1 byte 11, bit D6 'Mechanical Ventilation On' for determining whether system is mechanically ventilating or not.
47-49	Not Used	-	
50-52	Set oxygen concentration in fresh gas mode	% O ₂	
53-54	Not Used	-	
55-56	set P _{supp} set support pressure	cm H ₂ O, -	
57-58	Not Used	-	
59-61	Not Used	-	
62-64	set High RR alarm set High respiratory rate alarm limit	/min, -	
65-67	set Low RR alarm set low respiratory rate alarm limit	/min, -	
68-69	set apnea delay time	s, -	
70-71	Not Used	-	
72-73	Not Used	-	
74	Not Used	-	
75-76	Not Used	-	
77-79	Not Used	-	
80	Set patient type	'a'	adult
		'p'	pediatric
		'b'	ideal body weight (IBW)
81-83	Not Used	-	

84-86	Not Used	-
87-89	set patient weight	kg, -
90-92	Not Used	-
93-102	Not Used	-
103-107	Not Used	-
108-109	Not Used	-
110-111	Not Used	-
112-114	Not Used	-
115-116	Not Used	-
117-119	set rise rate	1-10, aut=auto, -
120-122	Not Used	-
123-124	set flow trigger set flow threshold for triggering a mechanical SIMV and PSV inspiration	(L/min)*10, - Note: If setting greater than 2 digits, then value is truncated to 99.
125-126	Not Used	-
127-128	set trigger window set percent of breath period where threshold for triggering a mechanical SIMV and PSV inspiration is enabled	% breath period, -
129-130	set End of Breath set percent of peak inspiratory flow threshold for ending a mechanical SIMV and PSV inspiration	% peak inspiratory flow, -
131-134	set Tinsp set inspiratory time	s*10, -
135-136	Not Used	-
137-139	Not Used	-
140-141	set mech RR set mechanical respiration rate for spontaneous ventilation modes	/min, -

142-143	set PSV backup delay set time to wait since last PSV breath before switching from PSV ventilation mode to SIMV ventilation mode	S, -
144-146	set High etCO ₂ alarm set High end tidal CO ₂ alarm limit	% CO ₂ *10 (volume), -
147-149	set Low etCO ₂ alarm set low end tidal CO ₂ alarm limit	% CO ₂ *10 (volume), -
150-152	set High FiCO ₂ alarm set High Fi CO ₂ alarm limit	% CO ₂ *10 (volume), -
153-155	set High etAA alarm set High end tidal anesthetic agent alarm limit	% AA*10 (volume), -
156-158	set Low etAA alarm set low end tidal anesthetic agent alarm limit	% AA*10 (volume), -
159-161	set High FiAA alarm set High Fi anesthetic agent alarm limit	% AA*10 (volume), -
162-164	set Low FiAA alarm set low Fi anesthetic agent alarm limit	% AA*10 (volume), -
165-167	Not Used	-
168-170	Not Used	-
171-174	Set fresh gas flow	(l/min)*100
175	Set balance gas type	'a' = Air, 'n' = N ₂ O.
176-178	Not Used	-
179	Not Used	-
180	Set flow sensor type	'd' De-lite/D-lite+
		'p' Pedi-lite
		'i' internal (ventilator)
181		'c' Circle

	Set circuit type Set patient breathing circuit type	'a'	non-circle ACGO
182	Set gas control mode	'f' = fresh gas control	
183-185	Not Used	-	
186-188	Not Used	-	
189-191	Not Used	-	
192-195	Not Used	-	
196	setting data status	See section 3.5.5.1.	
197-200	Not Used	-	
201-219	alarm status bytes	See section 3.5.5.2.	
220	Checksum	See Software Interface section.	
221	Carriage return	<CR>	

3.5.5.1 Setting Data Status Bitmap

The setting data status byte has seven bits of data from D6 (MSB) to D0 (LSB).

Note that a bit is set to 1 for an active condition (true) and to 0 for an inactive condition (false).

Byte 196	
Bit	Alarm/Status
D0	0 = 10 second data refresh 1 = new setting data
D1	0 = 10 second data refresh 1 = new alarm data
D2	0 – Multi Step Maneuver Inactive 1 – Multi Step Maneuver Active
D3	0 – Single Step Maneuver Inactive 1 – Single Step Maneuver Active
D4	Reserved
D5	Reserved
D6	Reserved

3.5.5.2 Status Bytes Bitmaps

The status bytes a string of 19 bytes, starting from left (byte 1) to right (byte 19). Each byte has seven bits of data from D6 (MSB) to D0 (LSB).

Note that the status byte bit is set = 1 for an active condition and 0 for an inactive condition.

Byte 201	
Bit	Alarm/Status
D0	FiO2 High, Gas Cell, No MGAS
D1	FiO2 Low, Gas Cell, No MGAS
D2	Not Used
D3	Not Used
D4	Not Used
D5	Replace O ₂ Sensor
D6	Calibrate O ₂ Sensor

Byte 202	
Bit	Alarm/Status
D0	Ppeak High
D1	Ppeak Low. Leak?
D2	Not Used
D3	PEEP High. Blockage
D4	Negative Airway Pressure
D5	Not Used
D6	Not Used

Byte 203	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Volume Mode Only. No PEEP or PSV
D3	Not Used
D4	MGAS APNEA
D5	Inspiration Stopped
D6	Other Priority Alarms (for high priority alarms not assigned a unique bit)

Byte 204	
Bit	Alarm/Status
D0	MVexp Low
D1	MVexp High High
D2	TVexp Low
D3	Tvexp High
D4	TV not achieved
D5	Apnea, Volume
D6	Apnea > 120 seconds, volume or CO ₂

Byte 205	
Bit	Alarm/Status
D0	No Inspiratory Flow Sensor
D1	No Expiratory Flow Sensor
D2	Inspiratory Reverse Flow. Check Valves OK?
D3	Expiratory Reverse Flow. Check Valves OK?
D4	Check Flow Sensors
D5	Volume Sensors disagree
D6	System Leak?

Byte 206	
Bit	Alarm/Status
D0	Unable to drive bellows
D1	Circuit Leak
D2	Circuit Leak Silenced
D3	Turn Power Off and On for Self Tests
D4	Aux Gas Blocked
D5	O ₂ Supply Pressure Low
D6	Not Used

Byte 207	
Bit	Alarm/Status
D0	N ₂ O Supply Pressure Low
D1	Air Supply Pressure Low
D2	Ventilator has no Drive Gas
D3	etO ₂ Low
D4	etO ₂ High
D5	FiO ₂ Low
D6	FiO ₂ High

Byte 208	
Bit	Alarm/Status
D0	RR Low
D1	RR High
D2	Memory (EEPROM) Failure
D3	Apnea CO ₂
D4	etCO ₂ Low
D5	etCO ₂ High
D6	FiCO ₂ High. Absorbent OK?

Byte 209	
Bit	Alarm/Status
D0	etDES low, etISO low, etSEV low
D1	etDES high, etISO high, etSEV high
D2	fiDES low, fiISO low, fiSEV low
D3	fiDES high, fiISO high, fiSEV high
D4	AGSS Flow Too High
D5	AGSS Flow Too Low
D6	Replace Insp or Exp Flow Sensor

Byte 210	
Bit	Alarm/Status
D0	Plug in Power Cable
D1	No Battery Backup
D2	Not Used
D3	Plug in Power Cable. On Battery, System shutdown in <x min
D4	MGAS ANE_WARMING_UP (5-minute warming up)
D5	No Battery Backup, Battery Cell shorted or Sulphated.
D6	No Battery Backup, Current Fault

Byte 211	
Bit	Alarm/Status
D0	Volume and Apnea Monitoring OFF, ACGO
D1	Check Circuit Connections, ACGO
D2	MGAS WARMING_UP (2-minute warming up)
D3	Display Panel Controls Failure
D4	Volume and Apnea Monitoring OFF, AUX
D5	Volume Compensation Locked (volume ventilation mode)
D6	Mechanical Ventilation On

Byte 212	
Bit	Alarm/Status
D0	O2 Flush Stuck On/Off
D1	Apnea Detection ON
D2	Not Used
D3	Increase Low MV limit
D4	Not Used
D5	Standby ON (set when anesthesia system is not in therapy mode or when respiratory care ventilator is in standby)
D6	Service Calibrations Due

Byte 213	
Bit	Alarm/Status
D0	Internal Failure Prevents Normal Operation
D1	Not Used
D2	Not Used
D3	Not Used
D4	Mixer Fail
D5	Not Used
D6	Not Used

Byte 214	
Bit	Alarm/Status
D0	Cooling Fan needs Service. System OK? Cooling Fan Failed. May overheat.
D1	Not Used
D2	Internal Failure, System may shutdown.
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 215	
Bit	Alarm/Status
D0	Alt O2 On
D1	Air Only Mode
D2	Vent Failure
D3	Mechanical Ventilation Disabled
D4	Not Used
D5	Not Used
D6	Not Used

Byte 216	
Bit	Alarm/Status
D0	Gas Module Failure, No CO ₂ , AA, O ₂ data
D1	Check Sample Gas Outlet
D2	Replace D-Fend
D3	Sample Line Blocked
D4	Check D-Fend
D5	Not Used
D6	Gas Module Not Compatible

Byte 217	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 218	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 219	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Backup Mode Active
D4	Not Used
D5	Not Used
D6	Not Used

3.5.6 Setup Data Response

Only returned when a request for setup data is received.

Each entry is zero filled and right justified – i.e. 95 expressed in four bytes is 0095.

‘-’ filled means data not available due to system state or configuration.

Byte(s)	Setup Data Name	Units and/or Description	
0-3	Header	:VTm	
4-5	Software revision number	01-99	
6-7	Service Pack Number	00-99	
8	Language	'0'=English, '1'=Spanish, '2'=German, '3'=Kanji, '4'=Dutch, '5'=Swedish, '6'=French, '7'=Italian, '8'=Danish, '9'=Polish, 'A'=Chinese, 'B'=Czech, 'C'=Hungarian, 'D'=Russian, 'E'=Portuguese, 'F'=Other, 'G'=Greek, 'H'=Finnish, 'I'=Turkish, 'J'=Norwegian, 'K'=Brazilian Portuguese	
9-10	Reserved		
11	Alarm loudness Alarm volume setting	No units, -	
12-13	Altitude setting	-4 to 36 in 100's of meters	
14	Drive gas	'O'	Oxygen
		'A'	Air
15	Model number	'g' = Carestation 750 Anesthesia	
16-30	System serial number	(limited to 7-bit ASCII characters)	
31	Checksum	See Software Interface section.	
32	Carriage return	<CR>	

3.5.7 Waveform Data Response

If Waveform Data Mode is enabled, a Waveform Data Response will be transmitted every 400 mS. Six blocks of ten data samples taken every 40 mS will be sent with each message. Each data value is a 3-digit, zero filled, right justified ASCII Hex representation of a 12 bit binary value. If a block is requested with an invalid character, the waveform data values for that block are filled with zeroes.

Each entry is zero filled and right justified – i.e. 95 expressed in three bytes is 095.

Waveform Data Response always contains six waveform blocks (unlike Ohmeda Com 1.0).

If the requested waveform data is not available, data is zeroed.

Byte(s)	Waveform Data Name	Units and/or Description
0-3	Header	:VTw
4-6	Waveform 1 1 st 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
7-9	Waveform 1 2 nd 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
.	.	.
.	.	.
.	.	.
28-30	Waveform 1 9 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
31-33	Waveform 1 10 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
34-36	Waveform 2 1 st 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
37-39	Waveform 2 2 nd 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
.	.	.
.	.	.
.	.	.
58-60	Waveform 2 9 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
61-63	Waveform 2 10 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
64-66	Waveform 3 1 st 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
67-69	Waveform 3 2 nd 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
.	.	.
.	.	.
.	.	.

88-90	Waveform 3 9 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
91-93	Waveform 3 10 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
94-96	Waveform 4 1 st 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
97-99	Waveform 4 2 nd 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
.	.	.
.	.	.
.	.	.
118-120	Waveform 4 9 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
121-123	Waveform 4 10 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
124-126	Waveform 5 1 st 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
127-129	Waveform 5 2 nd 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
.	.	.
.	.	.
.	.	.
148-150	Waveform 5 9 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
151-153	Waveform 5 10 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
154-156	Waveform 6 1 st 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
157-159	Waveform 6 2 nd 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)

.	.	.
.	.	.
.	.	.
178-180	Waveform 6 9 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
181-183	Waveform 6 10 th 40 ms waveform sample	hexadecimal "000" – "FFF" (see section 3.5.7.1 Waveform Data Description)
184	breath end/start index index to waveform sample at start of spirometry loop	0-9, -
185	Checksum	See Software Interface section.
186	Carriage return	<CR>

3.5.7.1 Waveform Data Description

Each waveform data sample shall be scaled as follows:

Pressure		
Range: -20 to 120 cmH ₂ O		
Raw Data	Scaled value	Transmitted
-20	0	"000"
0	512	"200"
120	3584	"E00"

Flow		
Range: -100 to 100 L/min		
Raw Data	Scaled value	Transmitted
-100	512	"200"
0	2048	"800"
100	3584	"E00"

Volume		
Range: 0 to 2000 mL		
Raw Data	Scaled value	Transmitted
0	512	"200"
2000	3584	"E00"

Anesthetic Agent		
Range: 0 – 20 %		
Raw Data	Scaled value	Transmitted
0	512	"200"
20	3584	"E00"

CO ₂		
Range: 0 – 35 %		
Raw Data	Scaled value	Transmitted
0	512	"200"
35	3584	"E00"

O ₂		
Range: 0 – 100 %		
Raw Data	Scaled value	Transmitted
0	512	"200"
100	3584	"E00"

3.5.8 Test Data Response

In auto mode, the Test Data Response will be transmitted every 1 minute.

Each entry is zero filled and right justified – i.e. 95 expressed in four bytes is 0095.

'-' filled means data not available due to system state or configuration.

Byte(s)	Setup Data Name	Units and/or Description
0-3	Header	:VTt
4-11	Current date stamp (yyyymmdd)	Year, month, day (example: 20031231), -
12-15	current time stamp (hhmm)	hour (24 hour format), minutes (example: 2359), -
16-17	current time stamp (ss)	seconds, -
18-21	Not Used	-
22-24	Not Used	-
25-27	Not Used	-
28-30	Not Used	-
31-33	Not Used	-
34-36	Not Used	-
37-39	Not Used	-
40-47	Circuit check date (yyyymmdd)	year, month, day (example: 20031231), -
48-51	Circuit check time (hhmm)	hour (24 hour format), minutes (example: 2359), -
52-59	System checkout date (yyyymmdd)	year, month, day (example: 20031231), -
60-63	System checkout time (hhmm)	hour (24 hour format), minutes (example: 2359), -
64-71	Gas Control check date (yyyymmdd)	year, month, day (example: 20031231), -
72-75	Gas Control check time (hhmm)	hour (24 hour format), minutes (example: 2359), -
76-83	Compliance check date (yyyymmdd)	year, month, day (example: 20031231), -
84-87	Compliance check time (hhmm)	hour (24 hour format), minutes (example: 2359), -
88-95	Not Used	-
96-99	Not Used	-

100-107	Not Used	-
108-111	Not Used	-
112-119	Not Used	-
120-123	Not Used	-
124-131	Not Used	-
132-135	Not Used	-
136-143	Not Used	-
144-147	Not Used	-
148-155	Not Used	-
156-159	Not Used	-
160-162	Des consump case measured Desflurane anesthetic agent consumption, cumulative since beginning of case	mL, ?, -
163-165	Not Used	mL, ?, -
166-168	Iso consump case measured Isoflurane anesthetic agent consumption, cumulative since beginning of case	mL, ?, -
169-171	Not Used	mL, ?, -
172-174	Sevo consump case measured Sevoflurane anesthetic agent consumption, cumulative since beginning of case	mL, ?, -
175-181	O ₂ consump case measured oxygen consumption, cumulative since beginning of case	L, ?, -
182-188	N ₂ O consump case measured N ₂ O consumption cumulative since beginning of case	L, ?, -

189-195	air consump case measured air consumption, cumulative since beginning of case	L, ?, -
196-199	Not Used	-
200-209	test status bytes	See section 3.5.8.1 Test Status Bytes Bitmaps .
210	Checksum	See Software Interface section.
211	Carriage return	<CR>

3.5.8.1 Test Status Bitmaps

The test status bytes are a string of 10 bytes, starting from left (byte 1) to the right (byte 10). Each byte has seven bits of data from D6 (MSB) to D0 (LSB).

Note that the bit is set to 1 for an active condition (true) and to 0 for an inactive condition (false).

Byte 200	
Bit	Alarm/Status
D0	Quick Check Result
D1	00 – None 01 – Pass 10 – Fail 11 – Abort
D2	Vent Check Result
D3	00 – None 01 – Pass 10 – Fail 11 – Abort
D4	Mixer Check Result
D5	00 – None 01 – Pass 10 – Fail 11 – Abort
D6	Not Used

Byte 201	
Bit	Alarm/Status
D0	Compliance Check Result
D1	00 – None 01 – Pass 10 – Fail 11 – Abort
D2	Leak Check Result
D3	00 – None 01 – Pass 10 – Fail 11 – Abort
D4	Agent Delivery Check Result
D5	00 – None 01 – Pass 10 – Fail 11 – Abort
D6	Not Used

Byte 202	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 203	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 204	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 205	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 206	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 207	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 208	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

Byte 209	
Bit	Alarm/Status
D0	Not Used
D1	Not Used
D2	Not Used
D3	Not Used
D4	Not Used
D5	Not Used
D6	Not Used

3.6 Example

The Datex-Ohmeda Com 1.0 serial protocol default is “1.0 Mode” protocol in “Slave Mode” with “Checksum Enabled”. So data **will not** automatically be sent out the serial port.

Upon power up or a reset, the Datex-Ohmeda anesthesia system device communicates in the Ohmeda Com 1.0 protocol mode. This allows compatibility with older external communications controllers. If an external communications controller wants to communicate using the Datex-Ohmeda Com 1.3 protocol mode, it must send the <ESC>VTO12c<CR> packet (“Select Datex-Ohmeda Com Protocol”). The Datex-Ohmeda Com 1.2 protocol provides addition information which is not available in the Ohmeda Com 1.0 protocol.

If an external communications controller wants to stay in “Slave Mode” it must send the <ESC>VT?c<CR> packet (“Send All Data”) to receive one Status Data and one Measured Data packet. The external communications controller will then need to request sending all the data on a periodic basis. This method is not recommended because the external communications controller has the potential to request sending all the data faster than the Datex-Ohmeda device is capable of responding.

So, the preferred method is to use “Auto Mode” by sending the <ESC>VTXc<CR> packet. This will allow the Datex-Ohmeda device to send the measured data and status data whenever changes occur, or every 10 seconds when changes do not occur. Using this method **will not** put a strain on the Datex-Ohmeda device and again is the preferred method to use.

Listed below is an example to get a Datex-Ohmeda device into Datex-Ohmeda Com 1.2 auto mode with no checksum and pressure waveform output enabled. Using this method the “c” [checksum] can be any character (including carriage return, <CR>) for all commands sent after the <ESC>VTDw<CR> command (in this example space, <SP>, is used).

<ESC>VTDw<CR>

<ESC>VTO12<SP><CR>

<ESC>VTX<SP><CR>

<ESC>VTwP00000<SP><CR>

- Disable checksum (“w” or 0x77 is the checksum value for this command)
- Select Datex-Ohmeda Com 1.2 protocol
- Set into Auto mode (<SP> used for checksum since checksum is disabled)
- Request pressure waveform data, no 2nd – 6th waveform data

4. Change History

Revision	Date	Reason For Change
1	22 January 2020	Initial creation for Carestation 750 Anesthesia.